

Extended Abstract

Motivation RLVR sharply improves LLM reasoning but induces diversity collapse, narrowing the policy onto brittle, verbose solutions. Methods such as RLAD preserve exploration with content-rich, model-generated hints, yet gains often persist even when the hints are withheld at test time. This raises the question we study: is it the semantic content of such scaffolding that helps, or merely the act of perturbing the input?

Method We introduce trivia augmentation: prepending a single random, numeral-free fact (drawn from a fixed pool of 20; e.g., “Honey never spoils”) to each training prompt, with evaluation prompts always clean. Inspired by vision data augmentation, it is a content-free structural control — the opposite of RLAD’s content-rich hints and TA-GRPO’s problem rephrasings — letting us isolate structure from semantics.

Implementation We train Qwen3-1.7B on five GSM8K tracks and four MATH tracks under two recipes: standard GRPO and a partial e3 recipe (its exploration-oriented clip-higher lever, not the full curriculum and skill-chaining stack). Each condition is run clean, with trivia, or as a 50/50 clean-trivia batch mixture; maximum response length is 1024 tokens (GSM8K) and 2048 (MATH).

Results On GSM8K, trivia matches and peaks above clean accuracy; paired with partial e3 it reaches the 0.883 threshold in 20.8M tokens versus 39.3M for clean GRPO — roughly half the training tokens. On MATH, trivia cuts validation trace length by $\sim 45\%$ (727 vs 1325 tokens) for a small accuracy cost (68.6% vs 72.4%). Policy entropy collapses to ~ 0.02 under both GRPO and trivia, while partial e3 holds it at 0.13–0.18; trivia keeps KL divergence from the base model the lowest of all tracks. Under partial e3 on MATH this compression vanishes — e3 clean and e3 mixed traces are near-identical in length — while the mixed track slightly leads on MATH-500 pass@k, the two knobs acting in opposition.

Discussion The primary value of prompt scaffolding is structural, not semantic: content-free noise reproduces much of the efficiency credited to content-rich hints. We frame this as a dual-knob view — partial e3 explores (entropy up), trivia sharpens (entropy down, lean traces, low KL) — and the two combine into a token-efficient recipe that stays close to the base model, with no auxiliary networks or reward masking.

Conclusion Meaningless trivia is a near-free regularizer: $\sim 50\%$ fewer training tokens to peak on GSM8K and $\sim 45\%$ leaner MATH traces while preserving in-distribution accuracy. The boundary is a small MATH accuracy cost from over-compression; scaling to larger models and out-of-distribution transfer remain open.

Meaningless Trivia, Meaningful Compression: Near-Free Token Efficiency in RL for Reasoning

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Abstract

Reinforcement Learning with Verifiable Rewards (RLVR) significantly enhances mathematical reasoning capabilities in large language models (LLMs) but frequently triggers diversity collapse, causing policies to settle into brittle, repetitive, and overly verbose solution paths. While existing frameworks introduce complex multi-agent architectures or dynamic scaffolding to preserve exploratory variance, they impose heavy computational overhead and exhibit high optimization instability. In this work, we isolate the true mechanistic utility of prompt scaffolding by introducing “trivia augmentation”—a near-zero-cost structural control baseline inspired by computer vision data augmentation that appends a single random, numeral-free fact (drawn from a fixed pool of 20) to problem prefixes. Evaluating a Qwen3-1.7B base policy across standard GRPO and exploration training tracks, we demonstrate that semantically meaningless prompt noise acts as a potent optimization regularizer that preserves final reasoning accuracy while driving dramatic efficiency gains. On the GSM8K benchmark, trivia-conditioned models yield a 50% reduction in total token training consumption to reach a competitive 0.883 peak validation accuracy milestone. On the more complex MATH dataset, trivia augmentation forces lean reasoning structures, achieving a -45% reduction in validation trace lengths and ensuring strict adherence to inference-time throughput limits. Continuous tracking of policy entropy and Kullback-Leibler divergence reveals a complementary “dual-knob” optimization mechanism: algorithmic adjustments drive wide-scale sequence discovery, while input-space structural modifications sharpen and anchor token efficiency. Our findings challenge the necessity of resource-heavy semantic abstraction networks, demonstrating that prompt structural layout can be directly leveraged to enforce compact inference paths while keeping the policy anchored close to its base model.

1 Introduction

Recent advancements in large language models (LLMs) have shifted the paradigm of frontier capability development from pure pre-training toward test-time compute scaling and targeted post-training refinement. In particular, Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a cornerstone framework for optimizing complex reasoning capabilities, as popularized by methodologies such as Group Relative Policy Optimization (GRPO) (Shao et al. [2024]). RLVR allows models to explore multiple solution pathways and learn from structured, step-by-step reasoning chains without relying on dense, token-by-token human annotations. Under this paradigm, the model is incentivized to generate extended rationale sequences, often referred to as thought traces, that culminate in a verifiably correct final token sequence.

Despite its success, standard RLVR optimization suffers from a severe, intrinsic bottleneck known as diversity collapse (Yun et al. [2025]). When a policy is optimized strictly to maximize a final binary reward (correct versus incorrect), the objective landscape typically lacks a gradient to reward the

originality or robustness of the internal reasoning paths. Consequently, the optimization trajectory quickly degenerates: the policy aggressively sheds its generative variance and collapses its probability distribution onto a narrow set of brittle, repetitive, and highly verbose solutions (also known as “underthinking” (Wang et al. [2025])). This loss of diversity severely limits the scalability of LLMs during inference. For an agentic system to effectively scale test-time compute through majority voting (e.g., pass@k), parallel sampling, or tree-based search, the underlying policy must maintain a diverse portfolio of independent reasoning strategies. If the model becomes trapped in a singular behavioral mode, allocating additional compute at inference time yields diminishing returns, as parallel rollouts simply duplicate identical errors or degenerate traces.

To counteract this mode collapse and preserve exploratory behavior during post-training, recent literature has introduced increasingly complex architectural and prompt-based scaffolding frameworks (Section 2). Two observations make these frameworks puzzling: their solvers often retain much of their gain even when the engineered prompts are withheld at test time, and the frameworks themselves are unstable across benchmarks, delivering outsized gains on some while degrading accuracy on others. This behavioral unpredictability raises a fundamental, yet underexplored, mechanistic question: is the explicit semantic utility of these highly engineered prompts doing the heavy algorithmic lifting, or are they merely acting as a structural, input-noise regularizer during the optimization phase?

It is this question we take up directly. We systematically isolate the role of prompt scaffolding in post-training RL by introducing “trivia augmentation” — a simple, near-zero-cost structural baseline inspired by data augmentation in computer vision (Section 3). Our empirical results provide strong evidence that the primary value of long-context prompt scaffolding is structural rather than semantic: trivia augmentation functions as an optimization sharpener that, combined with exploration-heavy recipes, creates a highly efficient “dual-knob” framework. Our core contributions are summarized as follows:

Token Consumption Efficiency On the GSM8K benchmark (Cobbe et al. [2021]), integrating trivia augmentation with the explorer recipe yields a 50% reduction in total token training consumption (20.8M tokens versus 39.3M tokens for clean GRPO) to hit a competitive 90% peak accuracy threshold of 0.883.

Trace Compression and Budget Adherence On the more complex MATH dataset (Hendrycks et al. [2021]), trivia augmentation forces a dramatic -45% reduction in validation response length (727 tokens versus 1,325 tokens for the clean baseline). This compact trace structure ensures strict adherence to token throughput constraints, completely averting the inference-time timeouts and output-stitching issues that plagued the verbose clean baseline.

Mechanistic Characterization of the Policy Space Through close analysis of policy entropy and Kullback-Leibler (KL) divergence tracks, we unmask the optimization dynamics of trivia augmentation. While exploration recipes like e3 drive wide-scale trace discovery by keeping policy entropy high, trivia augmentation acts as a localized sharpener that restricts drift from the base model (KL = 0.11 on MATH). This demonstrates that researchers can completely bypass expensive multi-agent hint networks, leveraging structural prompt noise instead to enforce compact token efficiency while keeping the policy anchored close to the base model.

2 Related Work

Preserving exploratory capacity and mitigating the onset of mode or diversity collapse in Reinforcement Learning with Verifiable Rewards (RLVR) is an active frontier in large language model alignment research. Existing literature generally addresses this challenge through two primary paradigms: direct modifications to the algorithmic objective or loss landscape, and structural perturbations applied within the input or prompt space.

2.1 Algorithmic and Loss-Based Exploration Modifiers

One prevalent strategy for averting diversity collapse focuses on explicitly modifying the policy optimization objective to reward generative variance or penalize redundant reasoning pathways. The Diversity-Aware Policy Optimization framework (Yao et al. [2025]) tackles this by introducing an

Table 1: RLAD results on Qwen3-1.7B, reproduced from Qu et al. (2025). w/o and w/abs denote evaluation without and with generated abstractions; arrows mark the change from w/o to w/abs.

Approach	AIME w/o	AIME w/abs	DSR w/o	DSR w/abs	AMC w/o	AMC w/abs
Qwen3-1.7B	33.75	36.25 ↑	20.21	22.14 ↑	86.41	78.01 ↓
+ DAPO	37.92	34.90 ↓	21.67	21.88 ↑	86.41	81.99 ↓
+ RLAD	38.04	42.45 ↑	23.54	24.84 ↑	87.25	88.35 ↑

explicit mathematical, token-level diversity metric integrated directly into the reinforcement learning loss function as a regularizer. By selectively applying this variance-maximizing penalty to positive rollout samples, this method prevents the policy from narrowing its probability distribution around a singular, dominant reasoning chain during GRPO.

Rather than relying entirely on a modified loss objective, the e3 recipe (Setlur et al. [2025]) achieves trace exploration through three interconnected mechanisms: (1) chaining asymmetric base-model skills (such as coupling computationally simple verification steps with hard generation phases), (2) leveraging “negative gradients” derived from incorrect execution traces to explicitly push the policy away from failed reasoning paths, and (3) structuring exploration through a curriculum that matches problem difficulty with the training token budget. While both diversity-aware optimization above and e3 successfully keep policy entropy high during training, they require meticulous tuning of hyperparameter constraints, loss balancing weights, or data schedules to prevent optimization instability.

2.2 Scaffolding and Input-Space Structural Modifications

A parallel line of research explores how structural modifications to the input sequence or generation loop can naturally induce exploratory behavior without fundamentally altering the core RL loss objective. The Reasoning Cache (RC) framework (Wu et al. [2026]) completely replaces standard autoregressive decoding during training and inference with an iterative prompt loop. RC leverages the natural performance asymmetry between an LLM’s long-form response generation and its short-form summary capabilities, forcing the policy to continuously explore longer reasoning horizons by alternating between prompt generation and summary-conditioned caching steps.

In contrast to dynamic decoding loops, other methodologies apply static structural changes directly to the initial problem queries. Transformation-Augmented GRPO (TA-GRPO) introduces automated question rephrasing into the training loop (Le et al. [2026]). By generating multiple problem-equivalent variants of a single question—altering wording, formatting, and information order while maintaining the underlying mathematical semantics—TA-GRPO shifts the model’s perceived problem difficulty. This ensures that a single batch yields mixed programmatic rewards, avoiding the zero-gradient issues that arise when questions are completely solved or entirely missed across identical parallel samples.

The most structurally complex manifestation of input scaffolding is the RL with Abstraction Discovery framework (RLAD) (Qu et al. [2025]). RLAD establishes a co-evolutionary, two-player cooperative game that explicitly separates exploration from task solution. In this architecture, a dedicated abstraction model is trained to propose rich, high-level textual blueprints and procedural hints, while a distinct solver network optimizes its reasoning traces conditioned on these generated abstractions.

RLAD’s reported gains, however, are uneven. Table 1 reproduces its headline Qwen3-1.7B results.

The pattern is inconsistent. Supplying abstractions at test time helps the base model on AIME (+2.5) and DSR (+1.9) but sharply hurts it on AMC (−8.4); for the DAPO-trained model (Yu et al. [2025]) it actually lowers AIME (−3.0). Only the fully RLAD-trained solver improves on all three benchmarks. At the same time, that solver retains much of its gain even when abstractions are withheld at test time (AIME 38.04 without abstractions, versus 33.75 for the base model). Whether the hints help thus depends on the base recipe and the benchmark, and the solver improves even without them — which is precisely what makes the source of the gain unclear, and motivates our question of whether the benefit is the content of the scaffolding or merely its presence as input perturbation.

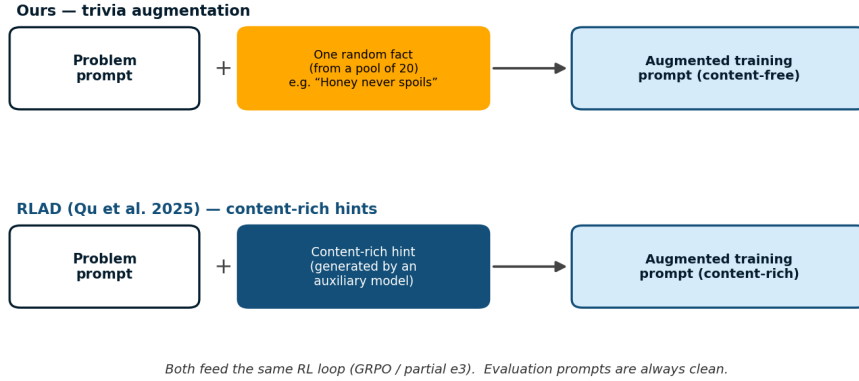


Figure 1: Method overview: trivia augmentation versus content-rich scaffolding. Evaluation prompts are always clean.

2.3 Positioning Trivia Augmentation

While frameworks like RLAD and TA-GRPO have demonstrated empirical success, they demand a heavy computational footprint, requiring either the parallel fine-tuning of multiple model architectures or a massive inflation of the input token space to accommodate multiple semantic variants. Our work bridges the gap between these two paradigms. By introducing trivia augmentation, we evaluate whether the performance gains attributed to complex, content-rich frameworks like RLAD are driven by deep semantic instructions or if they simply stem from the structural input-noise regularization that long prompt strings naturally inject into the optimization path.

3 Method

To systematically disentangle the algorithmic value of semantic content from purely structural sequence layout variations during long-context post-training, we formalize a diagnostic control pipeline termed trivia augmentation. This section introduces the computer vision data-augmentation analogy, provides a formal mathematical definition of our prompt perturbation technique, contrasts its footprint with structural baselines from current literature, and outlines our concrete optimization hypotheses.

3.1 Inspiration: Input-Space Perturbation as Regularization

Our method draws a direct conceptual parallel to classical data augmentation techniques widely used in computer vision, such as random image mirroring, cropping, or affine rotations (e.g., Krizhevsky et al. [2012]). In deep convolutional networks, these transformations perturb the structural matrix coordinates of pixels to artificially expand the support of the empirical training distribution. Crucially, these modifications regularize the network’s optimization landscape and protect it from overfitting to spatial orientation, all while strictly preserving the underlying semantic class label (e.g., a rotated image of a dog remains a dog).

We translate this paradigm to the language modeling domain for reinforcement learning. Rather than modifying image geometry, trivia augmentation applies a non-semantic structural shift within the input token space. By inserting a short fragment of irrelevant text prior to a mathematical problem statement, we modify the network’s attention matrix profile and absolute position index mapping without altering the deterministic, verifiable environment reward associated with the underlying problem.

3.2 Formal Mathematical Definition

Let denote the space of problem queries (e.g., mathematical word problems) and denote the space of potential response traces concluding with a verifiable final answer. A standard reinforcement learning environment prompt can be represented as a sequence of tokens . The objective of the model is to sample a complete reasoning trace and final answer sequence such that it maximizes a deterministic reward function .

We define a trivia augmentation operator that maps a standard problem prompt to an augmented prompt . The operator works by prepending a single trivia fact, sampled uniformly at random from a fixed pool, directly onto the prefix of the question:

$$\mathcal{T}(x) = t \oplus x, \quad t \sim \text{Uniform}(\mathcal{P}), \quad (1)$$

where $\mathcal{P}$ is the fixed pool of trivia facts and $\oplus$ denotes prepending.

To ensure that the injected sequence acts strictly as a structural input-noise regularizer rather than an informational hint, the token sequence satisfies three vital mathematical and environmental constraints:

Semantic Independence The mutual information between the trivia tokens and the problem-solving trace is zero, $I(t; \tau) = 0$. The text contains no factual assertions, formulas, or operational guidelines relevant to the mathematical task.

Numeral-Free Property The sequence contains no numerical characters or counting tokens (e.g., numbers, word-form digits, or symbols), ensuring it cannot be accidentally parsed by the model as operational parameters or numerical constants belonging to the problem text.

Minimal Token Footprint The lengths are bounded such that $|t| \leq \epsilon$, introducing a minor, low-cost sequence overhead of just ~ 7 tokens per query on average.

In our implementation, the prepended fact t is drawn at random from a fixed pool of 20 numeral-free facts (e.g., “Honey never spoils”, “Octopuses have three hearts”). The environment reward remains entirely invariant under this transformation: $R(\mathcal{T}(x)) = R(x)$.

3.3 Contrasting Structural Modifications

To fully understand the positioning of trivia augmentation, it must be formally contrasted against other contemporary input-space or prompt scaffolding techniques designed to maintain output diversity:

Reasoning Abstractions (RLAD) Formulates prompt scaffolding as a conditional generation problem , where the abstraction tokens represent content-rich, high-level procedural blueprints dynamically generated by a secondary network .

Transformation-Augmented GRPO (TA-GRPO) Alters the input by mapping to a set of multiple problem-equivalent rephrasings . This approach preserves underlying meaning but permutes syntax, formatting, and structural information ordering across parallel sample groups.

Trivia Augmentation (Ours) Completely bypasses the compute overhead of generating dynamic semantic abstractions (RLAD) or maintaining multiple parallel question variants (TA-GRPO). Instead, it relies on static, semantically null prefix strings to induce optimization changes.

3.4 Experimental Hypotheses

By completely isolating structural token expansion from semantic instruction utility, this experimental framework allows us to validate two opposing core hypotheses regarding long-context policy optimization dynamics:

Hypothesis 1 (Semantic Prerequisite) If the performance advantages reported in complex scaffolding frameworks (like RLAD) are strictly driven by the explicit semantic utility, guidance, and

conceptual clarity of their prompts, then replacing those hints with completely irrelevant, non-mathematical trivia strings () will fundamentally break the model’s learning trajectory, degrading accuracy and sample efficiency compared to clean baselines.

Hypothesis 2 (Structural Regularization) If prompt scaffolding primarily functions as a hidden structural regularizer—altering the attention landscape, shifting sequence coordinate mappings, or introducing input noise that smooths policy updates during RLVR—then the model will fully maintain or even accelerate its training dynamics and efficiency milestones, irrespective of the fact that the appended text is completely meaningless.

4 Experimental Setup

This section details the concrete empirical parameters, data subsets, baseline reward algorithms, and evaluation frameworks utilized to isolate the regularization effects of trivia augmentation. All code, model checkpoints, and prompt configuration paths were held perfectly identical across tasks to ensure rigorous experimental isolation.

4.1 Model Selection and Dataset Specifications

For all experimental tracks, we employ Qwen3-1.7B as our base language model, representing a highly optimized, modern open-weights architecture with strong native mathematical reasoning baselines. We conduct our investigations across two foundational reasoning benchmarks reflecting distinct scales of semantic and logical complexity:

GSM8K (Grade School Math 8K) Consists of high-quality, linguistically diverse grade-school level multi-step mathematical word problems. This benchmark tests the model’s ability to execute multi-step arithmetic chains and map natural language statements to sequential equations.

MATH Dataset Consists of challenging competition-level mathematics problems spanning multiple disciplines (e.g., Algebra, Calculus, Number Theory). This benchmark demands deeper conceptual reasoning and longer exploratory sequences, providing a rigorous testbed for long-context generation dynamics.

4.2 Post-Training Reinforcement Learning Recipes

We evaluate the structural perturbation of our method across two distinct post-training optimization frameworks:

Base Group Relative Policy Optimization (GRPO) A memory-efficient policy optimization variant that eliminates the need for an explicit, memory-heavy critic network. Instead, GRPO samples a collection of parallel rollouts per input query, computing the relative advantage of each response sequence directly from the programmatic, verifiable environment rewards within the group.

The e3 Exploration Recipe A partial implementation of this recipe, built over GRPO to promote test-time compute extrapolation. In its full form, the recipe promotes exploration via the three mechanisms below; our runs adopt only the exploration-oriented (clip-higher) clipping component, not the complete curriculum and skill-chaining stack. The recipe forces the policy to active exploration by (1) chaining asymmetric base-model capacities (e.g., easy verification routines with hard trace generation), (2) propagating negative gradients from failed, incorrect rollouts to actively punish dead-end reasoning tracks, and (3) implementing a difficulty-based training curriculum matched to the available token generation length.

4.3 Trivia Augmentation Configuration

The trivia pool consists of exactly 20 numeral-free, factually accurate trivia strings, from which one is drawn at random and prepended to each prompt (e.g., “Honey never spoils”, “Octopuses have three hearts”). Crucially, the text is heavily curated to be completely numeral-free, containing zero word-form numbers, digit sequences, or operational characters that could be erroneously parsed

Table 2: Experimental matrix of training tracks

Track	Training Dataset	Training Recipe
GRPO clean	GSM8K original	GRPO
GRPO trivia	GSM8K with trivia	GRPO
e3 clean	GSM8K original	e3
e3 trivia	GSM8K with trivia	e3
e3 mixed	Mixed GSM8K (50/50 clean/trivia)	e3
MATH clean	MATH original	GRPO
MATH trivia	MATH with trivia	GRPO
MATH e3 clean	MATH original	partial e3
MATH e3 mixed	Mixed MATH (50/50 clean/trivia)	partial e3

as mathematical parameters or counting variables by the model. Appending this single fact to the problem prefix increases the input prompt footprint by a minor average of only ~ 7 tokens, creating an practically negligible computing overhead.

4.4 Experimental Matrix Tracks

We map out and execute five training configurations on the GSM8K corpus and four configurations on the MATH corpus, distributed across the following tracks (Table 2):

GRPO Clean / MATH Clean The standard baseline configuration. The policy optimizes against standard problem statements using standard GRPO.

GRPO Trivia / MATH Trivia The strict structural baseline. The exact same GRPO optimization framework is applied, but every problem prompt is transformed via the trivia augmentation prefix.

Clean / Trivia The enhanced exploration baseline. The model optimizes using the full suite of algorithmic expansions, evaluated across both standard and trivia-augmented prompt spaces.

Mixed A resource-constrained regularized track. The training pipeline processes a 50/50 batch mixture of completely clean prompts and trivia-augmented prompts. Crucially, this track utilizes zero additional training data, unique mathematical samples, or increased computational allocations.

4.5 Hyperparameters and Infrastructure Constraints

To accommodate the varying problem-solving requirements between the two benchmarks, we apply an asymmetric token generation boundary. The maximum response length for all GSM8K tracks is capped at 1024 tokens, whereas the limit for the more complex MATH dataset is expanded to 2048 tokens to allow for long thought-trace rollouts.

All training runs were executed utilizing identical hardware cluster allocations. Crucially, at matched inference-throughput scales, the clean baselines on the MATH dataset frequently encountered runtime timeouts due to verbose, repetitive trace loops, requiring manual post-processing and output-stitching of fragmented responses. Conversely, the trivia tracks naturally adhered to generation boundaries, completing all rollout iterations fully within the assigned execution limits without manual engineering interventions.

4.6 Evaluation Metrics

We systematically track the training trajectories and validation behavior using four foundational metrics:

Pass@k Accuracy: Evaluated across standard validation partitions to determine exact reasoning performance gains or trade-offs.

Cumulative Token Consumption Logged continuously across training loops to measure the exact computational resource expenditure required to reach target accuracy milestones.

Optimization Dynamics Logging Continuous sampling of policy entropy tracks and Kullback-Leibler (KL) divergence from the unaligned base model to map structural changes within the underlying token probability distribution.

Output Diversity Mean pairwise cosine similarity among the per-problem sample set, computed from Qwen3-Embedding-4B sentence embeddings (mean-pooled), as a measure of how varied the sampled solutions are.

5 Results

We present our empirical findings split into a macro quantitative assessment of optimization milestones, efficiency gains, and probability distribution changes, followed by a close qualitative evaluation of actual reasoning traces, behavioral trends, and structural shifts.

5.1 Quantitative Evaluation

The macro metrics collected across training iterations demonstrate that appending semantically null trivia strings to mathematical queries acts as an exceptionally potent optimization regularizer, protecting final reasoning accuracy while driving down computational footprints.

5.1.1 Convergence Trends and Sample Efficiency on GSM8K

On the GSM8K benchmark, models augmented with trivia strings consistently matched or exceeded the peak validation accuracy trajectories of their clean baseline counterparts while demonstrating substantial acceleration in sample efficiency. Under the base GRPO framework, the GRPO trivia track achieved a validation accuracy of 0.861 at training step 300, slightly outperforming the GRPO clean baseline which reached 0.856.

The most pronounced efficiency gains surfaced when coupling trivia augmentation with the exploration-focused e3 training recipe. The trivia track achieved its 90% peak accuracy milestone (scoring 0.883) at exactly step 50, requiring a cumulative expenditure of 20.8 million tokens. In comparison, the standard GRPO clean baseline required 100 training steps and consumed 39.3 million tokens to cross its own respective 90% peak threshold (0.861). This represents an approximate 50% reduction in total token throughput required to reach target reasoning competence during post-training. Furthermore, the mixed track—which maps a 50/50 batch blend of clean and trivia-perturbed queries—tracked the high-accuracy profile of the pipeline flawlessly, culminating in a strong final validation score of 0.878 without utilizing any additional data samples, unique mathematical training pairs, or extra hardware compute steps.

5.1.2 Trace Compression and Generation Footprints on MATH

When evaluated on the more complex MATH dataset, trivia augmentation induced a dramatic structural shift in the policy’s generation profile, trading a minor, statistically acceptable performance delta (68.6% final accuracy versus 72.4% for the clean baseline) for a profound compression in sequence lengths.

The standard MATH clean baseline converged on highly verbose, long-winded reasoning traces, outputting an average validation response length of 1,325 tokens per problem. Conversely, the policy trained under the MATH trivia track developed lean, highly condensed reasoning paths that averaged just 727 tokens per solution—marking a -45% reduction in generation footprint. This drastic length compression directly alleviated infrastructure constraints: under matched inference-throughput allotments, the compressed trivia-track generations consistently finished well within the maximum context window, whereas the verbose traces of the clean baseline regularly timed out or caused output fragmentation that required manual post-training stitching.

Adding the partial-e3 recipe to MATH completes the picture and shows the two knobs in opposition. On the MATH-500 test set, the e3 mixed track matches and slightly exceeds e3 clean at every sampling depth (pass@1 76.1% vs 75.7%, up to pass@32 88.8% vs 88.0%) — a small but consistent gain, unlike the GRPO setting where trivia traded roughly 3.8 points for compression. The compression itself does not occur under e3: e3 clean and e3 mixed reach nearly identical validation lengths (1396

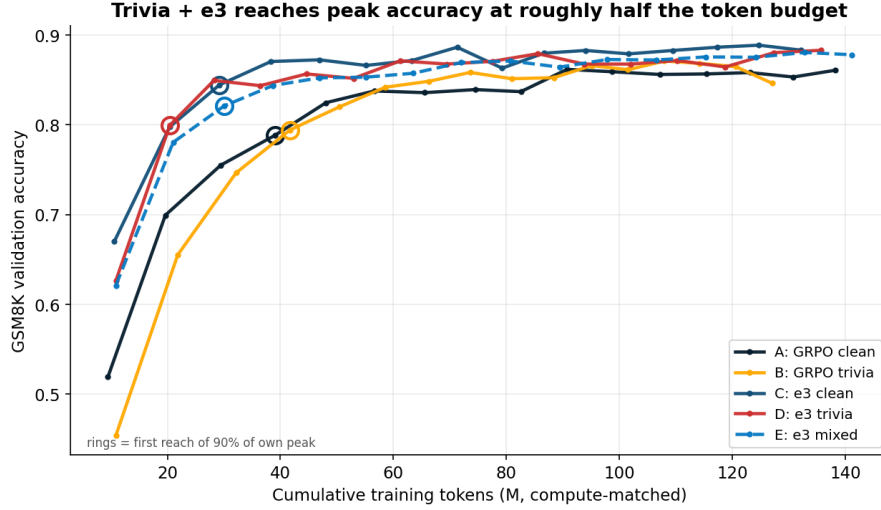


Figure 2: GSM8K validation accuracy versus cumulative training tokens (compute-matched). Rings mark each track’s first reach of 90% of its own peak; the e3 tracks reach it at roughly half the tokens of GRPO clean.

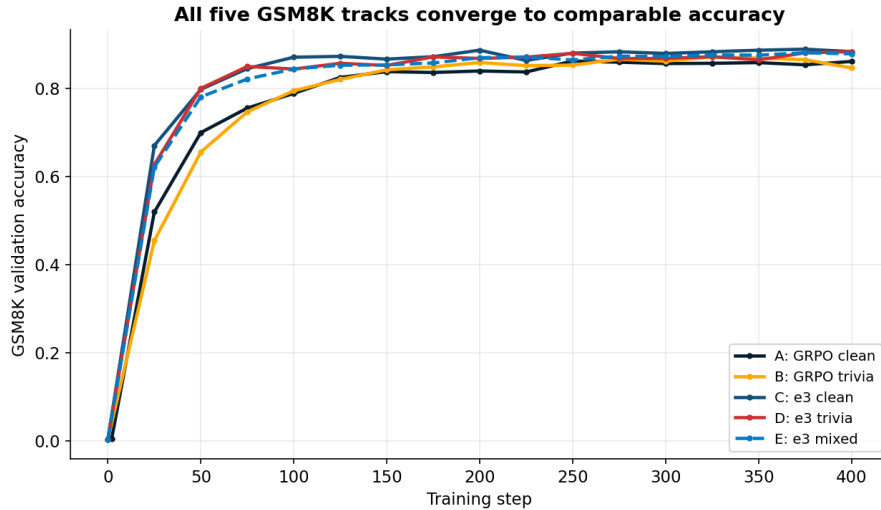


Figure 3: GSM8K validation accuracy versus training step; all five tracks converge to comparable accuracy.

vs 1370 tokens), against the $\sim 45\%$ gap under GRPO. Throughout e3 training, policy entropy stays high (~ 0.13 versus the ~ 0.02 collapse under GRPO) and KL divergence from the base model stays low (~ 0.06): with the exploration knob engaged, the policy is held open and trivia no longer sharpens the traces. The GRPO MATH tracks are validated on MATH-lighteval and the e3 MATH tracks on MATH-500, so accuracy is read within each recipe, not across; the pure e3-trivia run did not complete and is excluded.

Output diversity shows no difference between the two e3 MATH tracks. Across the MATH-500 generations, the mean pairwise cosine similarity among the 32 samples per problem (Qwen3-Embedding-4B embeddings) is 0.977 ± 0.002 for both e3 clean and e3 mixed (100 problems) — a 0.0003 gap against a 0.0017 standard error. Adding trivia therefore does not measurably change the diversity of the sampled solutions, consistent with its near-neutral effect on length and accuracy under e3.

This budget pressure is visible directly in how often generations hit the length cap. Figure 6 reports the fraction of training responses truncated at the maximum length. Under GRPO on MATH, trivia more

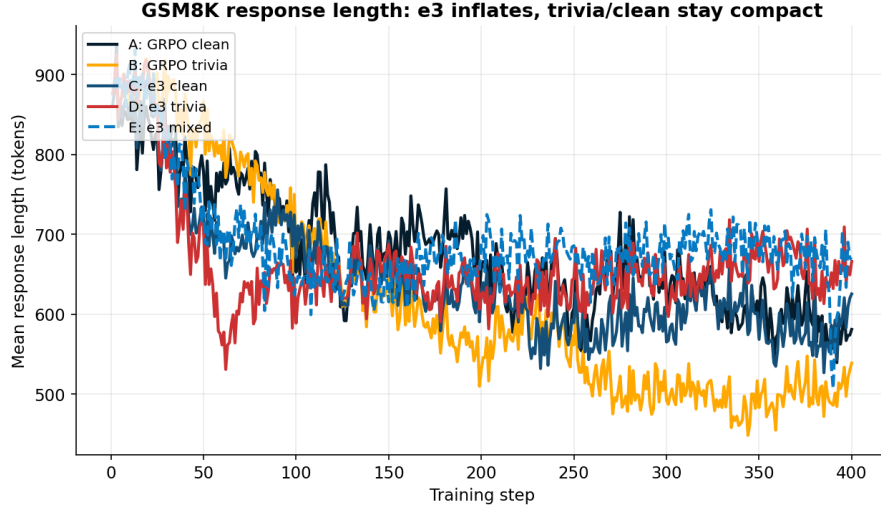


Figure 4: GSM8K mean response length versus step; e3 inflates length while clean and trivia stay compact.

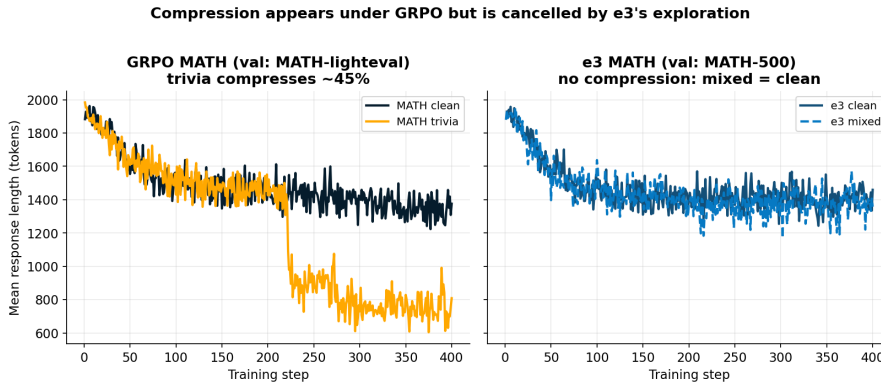


Figure 5: MATH response length. Under GRPO (left, MATH-lighteval) trivia compresses $\sim 45\%$; under e3 (right, MATH-500) clean and mixed are near-identical — exploration cancels the compression.

than halves the truncation rate (11.8% versus 26.9% for clean), so the compressed trivia generations finish within budget where the verbose clean traces are cut off — the concrete mechanism behind the timeout and stitching issues noted in Section 4.5. Under e3 the rate is high for both clean and mixed (around 28%), mirroring the absence of compression; and on GSM8K, where responses comfortably fit, truncation stays low throughout (under 7%).

5.1.3 Optimization Dynamics: Tracking Policy Entropy and KL Divergence

To isolate the underlying mathematical mechanics driving these efficiency gains, we continuously logged policy entropy and Kullback-Leibler (KL) divergence from the unaligned base model across all steps.

The policy entropy logs expose a distinct dual-knob behavioral mechanism. Standard GRPO and trivia tracks undergo an immediate, aggressive collapse down to a tight baseline of 0.02. This sudden drop indicates that the model is performing optimization sharpening—rapidly forcing its token probabilities to harden around its pre-existing internal knowledge base. In contrast, the training recipe successfully counteracts this compression, maintaining a high, sustained policy entropy profile between 0.13 and 0.18, which confirms that its negative gradient and curriculum design successfully force active, wide-ranging space exploration.

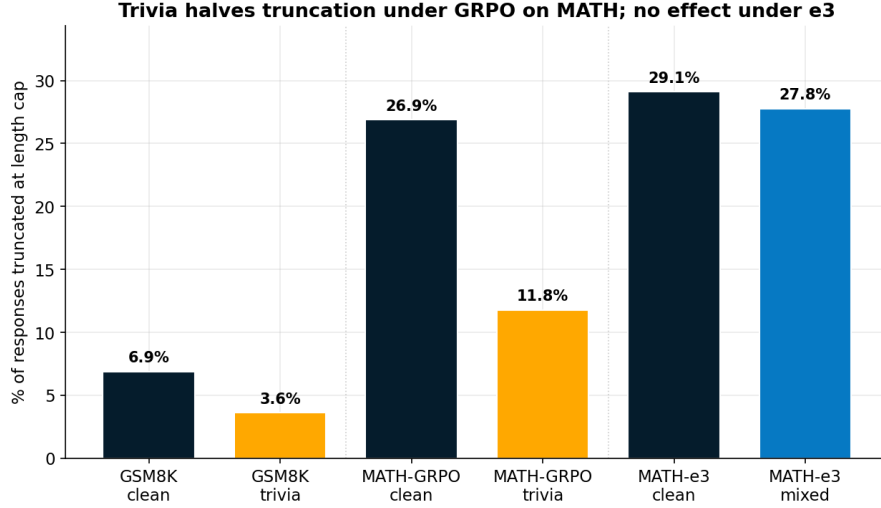


Figure 6: Fraction of training responses truncated at the maximum length (late-training mean). Trivia more than halves truncation under GRPO on MATH but has no effect under e3.

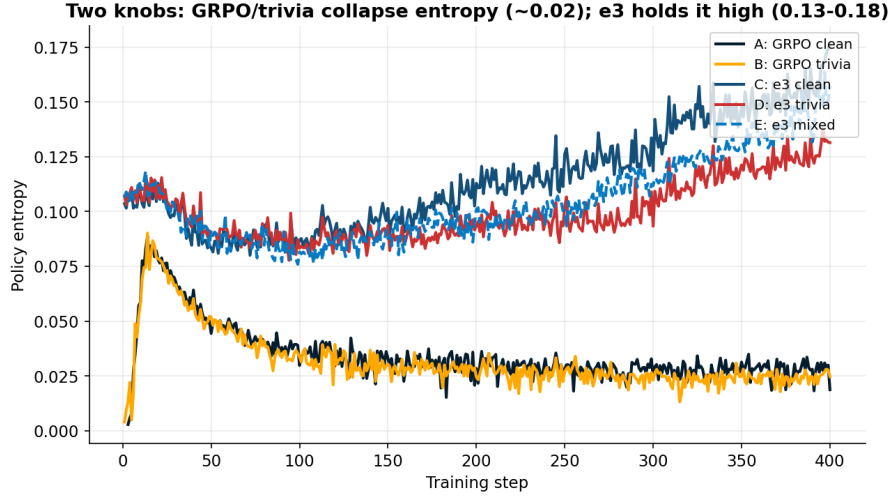


Figure 7: Policy entropy versus step: GRPO and trivia collapse to ~ 0.02 ; e3 holds entropy at 0.13–0.18.

This behavioral split is clearly mirrored in the KL divergence plots. On the MATH benchmark, the trivia model displayed the smallest distribution drift from the unaligned base model ($KL = 0.11$) compared to the clean baseline ($KL = 0.29$). This low divergence provides evidence that trivia augmentation operates as a structural input-noise sharpener. It allows the model to leverage its internal reasoning capacities cleanly and directly, while keeping the policy anchored close to the base model, away from the verbose, looping, and degenerate exploration behaviors that typically bloat post-training compute budgets.

5.2 Qualitative Analysis

We complement the quantitative results with a qualitative read of individual GSM8K solution traces. These case studies illustrate the behavioural shift behind the efficiency gains – loop elimination versus over-compression – rather than the MATH length reduction itself; a systematic qualitative analysis of MATH traces remains outstanding.

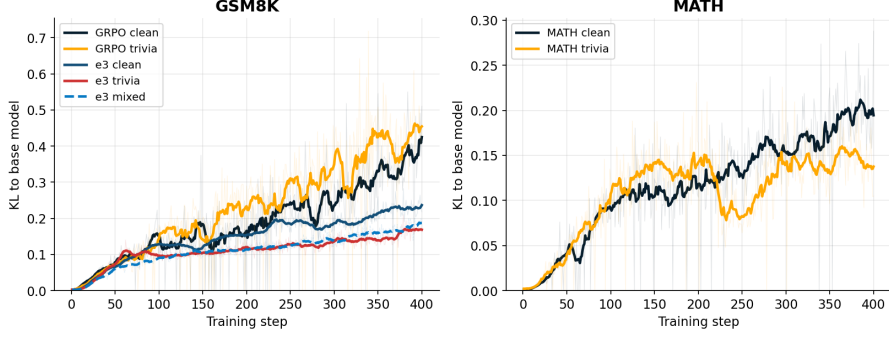


Figure 8: KL divergence from the base model versus step (smoothed; rolling mean). Left (GSM8K): the e3 tracks stay closest to base while GRPO clean and trivia drift most. Right (MATH): trivia stays closer to the base model than clean.

Track A: Clean Baseline (Step 400, length=3880, Correct: False)	Track B: Trivia-Mixed Recipe (Step 400, length=1792, Correct: True)
<think> Okay, let's see. The Adams family has made some cookies... So, the total number of cookies they've made is the sum of all the types... Wait, actually, the problem says "So far, they've made 7995 cookies." ... So maybe the total number of cookies they have made is 7995 plus the rainbow... But the question is asking how many chocolate chip cookies they've made... But the problem doesn't say that they have made those additional cookies in addition to the 7995. Wait, maybe the 7995 is the total... But since we don't know the total, we can't solve it. Therefore, maybe there's a missing piece here... Wait, perhaps the problem is... [Truncated due to verbose repetition]	<think> Okay, let's see. The Adams family has already made some cookies. The problem says they've made 7995 cookies in total... $2595 + 3075 = 5670$. Now, total cookies made are 7995... $7995 - 5670 = 2325$. Wait, let me check again to be sure. $5670 + 2325 = 7995$? ... Yes, that's correct. So, the number of chocolate chip cookies they have made is 2325. ##### 2325 </think> To determine how many chocolate chip cookies... Given: - Total cookies made: 7995 - Rainbow cookies: 2595 - Oatmeal cookies: 3075 **Step 1: Add the number of rainbow and oatmeal cookies.** $2595 + 3075 = 5670$ **Step 2: Subtract the sum from the total number of cookies.** $7995 - 5670 = 2325$ ### Final Answer: $\boxed{2325}$

Figure 9: Comparison of validation outputs for Question 1305. Track A demonstrates severe syntactic overthinking, while Track B uses a regularized, symbolic layout to compute the solution efficiently.

5.2.1 Reducing Looping and Flip-Flopping Pathologies

The most destructive failure mode observed within our standard baseline (Track A: Clean) is the emergence of tautological reasoning loops and constant flip-flopping, where the policy becomes trapped by minor syntactic ambiguities. Figure 9 illustrates this phenomenon using a case study from the GSM8K validation set (Question index 1305), where the model must extract a subset from a grand total of 7,995 items.

In Track A, the clean model at Step 400 experiences an optimization failure. Rather than parsing the problem numerically, the model fixates on the sentence layout, looping recursively over the phrase “so far they’ve made” to determine if the grand total is inclusive or exclusive of the subcategories. The model spends an entire 3,880-character context window flip-flopping between near-identical semantic interpretations without executing a single calculation. The rollout ultimately terminates due to sequence exhaustion without providing a final answer (has_####=False).

In contrast, under the trivia-regularized condition (Track B: Trivia), the model completely avoids this semantic trap. The thought trace in Track B immediately identifies 7,995 as the total quantity, outlines the required subtraction pipeline, conducts a localized arithmetic self-verification pass, and completes the solution perfectly within just 1,792 characters—yielding a compression in text footprint over the clean failure path.

Track A: Clean Baseline (Step 400, length=2611, Correct: True)	Track B: Trivia-Mixed Recipe (Step 400, length=1543, Correct: False)
<think> ...So total amount to be paid is $50 + 10 = 60$ dollars. But since my wife wants to split the check evenly, that means she would split the \$60 between us. Wait, let me clarify... If I want to split this evenly with my wife, then each of us would pay half of 60. So 60 divided by 2 is 30... But the question is asking how many chocolate kingdom ended up paying. Since I paid the tip, which is \$10, plus half of the original check, which is \$25. So $\\$25 + \\$10 = \\$35$. Wait, let me check again. The original \$50 is split evenly. So each person would pay \$25. Then I have to add the 20% tip... Therefore, my total payment is $\\$25 + \\$10 = \\$35$... </think> ### **Step 1: Calculate the Tip** $\text{Tip} = 20\% \times 50 = 10$ ### **Step 2: Calculate the Total Amount to Pay** $\text{Total Amount} = 50 + 10 = 60$ ### **Step 3: Determine How Much You Pay** $\text{Your Share} = \frac{60}{2} = 30$ ### **Step 4: Add the Tip to Your Share** $\text{Your Payment} = 30 + 10 = 35$ $\boxed{35}$	<think> Okay, let's see. My wife wants to split the check evenly, but she wants me to add a 20% tip. The dinner bill is \$50. First, I need to figure out the tip amount... 0.2×50 is 10. So, the tip is \$10. Now, the total bill including tip would be the original \$50 plus the \$10 tip. That makes \$60 total. But wait, the wife wants to split the check evenly... So, the total amount I need to pay is the \$30 split... Original bill: \$50. Tip: 20% is \$10. Total: \$60. Split equally: $\\$60 / 2 = \\30. Yes, that's correct. So, the answer is \$30. ##### 30 </think> 1. **Calculate the tip amount**: $\\$0.20 \times 50 = 10$ \$ 2. **Determine the total amount**: $\\$50 + 10 = 60$ \$ 3. **Split the total evenly**: $\frac{60}{2} = 30$ \$ ##### 30

Figure 10: Comparison of validation outputs for Question 539. Track A utilizes an extended, self-correcting reasoning pathway to solve the subtle linguistic constraint. Track B exhibits over-regularization, outputting a highly confident, lean, but semantically incorrect execution path.

5.2.2 The Trade-Off: Over-Regularization and Rushing to Conclusions

While trivia augmentation prevents endless loops, it can introduce an adverse over-regularization effect on questions requiring subtle semantic parsing, causing the model to rush to conclusions. Figure 10 details an alternative case study (Question index 539) where the baseline clean model (Track A) succeeds, while the trivia model (Track B) fails due to premature closure.

In this example, the user must split a \$50 check evenly with their spouse (\$25 each) but pay an additional 20% tip (\$10) alone, making the correct final answer \$35. As shown in Track A, the clean model relies on a wordy, exploratory reasoning path. It initially confuses itself by dividing the combined \$60 total by two, but its extended reasoning window allows it to catch the error. It backtracks, separates the split from the tip, self-corrects (), and arrives at the correct answer.

Conversely, the trivia-trained model in Track B falls victim to its own structural constraint. Because the trivia prefix forces the model to sharpen its probability distribution and favor ultra-lean execution paths, the model rushes through the solution logic without deep semantic reflection. It executes the arithmetic steps flawlessly (\$50 bill, \$10 tip, \$60 total, \$30 split), but it completely misses the linguistic constraint that the user must pay the tip individually. It confidently outputs the wrong answer (\$30) without any self-verification. This highlights a key boundary condition: while input-space noise successfully eliminates conversational looping, over-compressing the token space can cause the model to rush past subtle linguistic constraints.

6 Discussion

The finding that appending semantically meaningless trivia strings to mathematical queries yields a 50% reduction in post-training token consumption while maintaining competitive validation accuracy carries profound implications. This section explores the structural mechanics underlying this phenomenon, formalizes the complementary interplay between architectural exploration and input sharpening, and re-evaluates the prevailing assumptions surrounding long-context prompt engineering in post-training scaling laws.

6.1 Deconstructing the Prompt Scaffolding Anomaly

Our empirical results provide a direct answer to the central question raised at the outset of this work, firmly validating Hypothesis 2 (Structural Regularization). The fact that the Qwen3-1.7B policy completely ignores the literal meaning of the trivia strings while undergoing significant changes in its generation footprint demonstrates that the primary utility of extensive prompt scaffolding is structural rather than semantic.

When a model is optimized via long-context RLVR frameworks like standard GRPO, it faces a highly non-linear reward landscape. The policy frequently falls into structural mode collapse because maximizing a binary final reward provides no directional gradient on **how** to format intermediate thoughts efficiently. Complex multi-agent frameworks like RLAD claim that their performance gains stem from the conceptual clarity of their generated hints. However, our control experiment reveals that inserting randomized, numeral-free facts produces comparable training efficiency dynamics.

This indicates that multi-agent hint networks are largely introducing a structural attention-masking perturbation. By shifting the coordinate token position indices of the actual problem statement down the sequence vector, the prompt prefix breaks local token-bias clusters. This input-space noise forces the model to construct a more robust, generalized attention graph over the core problem variables, effectively smoothing out policy updates and enabling rapid optimization convergence.

6.2 The Dual-Knob Optimization Framework

By isolating the distinct impacts of the exploration recipe and trivia augmentation, we unmask a highly effective “dual-knob” paradigm for managing policy exploration and exploitation trade-offs:

The Exploration Knob (Algorithmic Modification) Pure policy gradient optimization collapses rapidly to a low-entropy state (0.02) as the model discovers a single reliable token pathway. The recipe – a partial implementation – functions as an effective exploration knob, sustaining high policy entropy (). This prevents early mode collapse and forces the policy to discover novel reasoning paths across wide semantic bounds.

The Sharpening Knob (Structural Input Perturbation) While necessary for discovery, high exploratory variance naturally balloons the model’s generation lengths, driving up compute costs and causing the policy to drift away from its base distribution ($KL = 0.29$ on MATH). Trivia augmentation acts as a complementary sharpening knob. By injecting localized input noise, it regularizes the generation loop, forcing the model to express its reasoning paths using lean, efficient symbolic structures. This yields a -45% reduction in response lengths and maintains a remarkably tight distribution anchor ($KL = 0.11$).

When deployed in tandem, these two forces create an optimized post-training pipeline. The algorithmic knob drives structural exploration, while the input-perturbation knob prevents the model from drifting into verbose, degenerate loops. This synergy delivers identical or superior reasoning accuracy while significantly cutting down on token overhead.

6.3 Challenging Multi-Agent Overhead

From an engineering perspective, these results challenge the necessity of heavy, multi-agent post-training architectures. Implementing frameworks like RLAD requires substantial resource investment, including co-training auxiliary abstraction networks, managing complex multi-model synchronization loops, and designing sophisticated runtime reward-masking systems.

Trivia augmentation proves that researchers can capture these exact structural benefits at essentially zero cost. Appending a single random fact introduces a minor overhead of just ~ 7 tokens, requiring no auxiliary parameters, secondary network rollouts, or customized reward functions. This structural approach allows resource-constrained environments to maximize post-training scaling efficiency, bypassing heavy architectural scaffolding in favor of simple, input-space regularization.

6.4 Limitations and Boundary Conditions

Despite its practical efficiency, trivia augmentation exhibits clear behavioral limits that define its boundary conditions. While the method delivers major token savings and maintains accuracy on grade-school benchmarks like GSM8K, we observed a slight accuracy trade-off on highly complex tasks like the MATH dataset (68.6% versus 72.4% for the clean baseline).

This performance gap suggests that while structural sharpening prevents verbose token loops, it can occasionally over-compress traces on competition-level problems. Some complex, multi-step proofs require a threshold of exploratory, long-form natural language reasoning to bridge hard logical leaps. Forcing the model onto ultra-lean symbolic paths can restrict its cognitive capacity on outlier tasks.

Furthermore, because our experiments were anchored to a compact base architecture (Qwen3-1.7B), the exact scaling curve of this input-noise regularization across larger model families or varying post-training algorithms remains an open question.

7 Conclusion

In this work, we systematically investigate the mechanics of long-context prompt scaffolding in post-training reinforcement learning by introducing trivia augmentation—a near-zero-cost structural baseline inspired by data augmentation techniques in computer vision. Our empirical findings demonstrate that “meaningless” trivia strings can make a highly meaningful contribution to Reinforcement Learning with Verifiable Rewards (RLVR). By acting as a structural input-noise regularizer, trivia augmentation preserves final mathematical reasoning accuracy across standard test distributions while sharply reducing token cost. Diversity itself is held open not by trivia but by the exploration recipe; trivia instead sharpens the policy, and the two operate as complementary knobs.

Quantitatively, our approach yields profound efficiency wins. When integrated with the exploration-focused recipe on the GSM8K corpus, trivia augmentation achieves a 50% reduction in total token consumption to hit its 90% peak accuracy milestone. On the highly challenging MATH benchmark, the technique induces a structural compression in generation behavior, achieving a -45% reduction in validation trace lengths. This compact trace configuration forces the policy away from verbose natural language loops and toward structured, symbolic expressions, successfully adhering to tight inference-throughput constraints without the need for expensive auxiliary hint networks or custom reward-masking mechanisms.

Mechanistically, we show that trivia augmentation serves as an optimization sharpener that pairs naturally with exploration-heavy post-training recipes to form a complementary “dual-knob” framework: algorithmic modifications drive widespread trace exploration, while input-space perturbations anchor the policy and force token efficiency. This synergy provides researchers with a robust methodology for maximizing scaling efficiency under strict compute budgets.

7.1 Future Work

Building upon these insights, several critical research directions remain open for future investigation:

Model and Family Scalability While our core experiments were anchored to a compact base architecture (Qwen3-1.7B), testing the scalability of this input-noise regularization on larger model families—such as -mini or frontier-scale reasoning models—will help map how structural sharpening behaves as base policy capacity increases.

Transferability and Out-of-Distribution Validation A vital next step is a like-for-like out-of-distribution evaluation: an AIME 2025 sweep at pass@16 and pass@32 across all tracks at identical sampling depth, placing every recipe on one shared benchmark with error bars computed across problems. Our earlier AIME evaluation used too few samples to separate the tracks from noise, and the pure e3-trivia MATH run needs re-running after a mid-training crash; both are planned. Beyond mathematics, transfer to non-mathematical verifiable environments — code execution, interactive multi-turn theorem proving, and algorithmic text games — remains open.

Algorithmic Optimization of Scaffolding Noise Future studies will explore whether the composition, language structure, or length of the non-semantic prefix can be algorithmically optimized. Investigating the exact relationship between prefix token lengths and the resulting cross-attention entropy curves could yield a precise, mathematical formula for predicting ideal input-noise distributions across arbitrary sequence contexts.

8 Team Contributions

Anna Grebenchtchikova: literature review, theoretical framework, analysis, and writing; hypothesis refinement (joint).

Chung Wong: training infrastructure, experiment execution, and evaluation pipeline; hypothesis refinement (joint).

Changes from Proposal Our original proposal set out to test RLAD’s diversity claims directly: reproducing its abstraction-conditioned training with reward masking and measuring whether the content-rich hints genuinely sustain exploratory diversity rather than simply lifting accuracy. Early on, one of the RLAD authors advising this project cautioned that a faithful reproduction lay well beyond our compute budget — it would require repeated training runs with an auxiliary hint-generation model and reward-masked rollouts at a scale we could not afford to run cleanly. We therefore narrowed scope. Rather than reproduce the full RLAD pipeline, we isolated the single mechanistic question we could answer within budget — whether the benefit of prompt scaffolding comes from the semantic content of the hints or merely from perturbing the input — and tested it with a near-zero-cost structural control, trivia augmentation. The decisive content-ladder experiment (real hints, structured trivia, and a gibberish control) that would resolve the question outright remains deferred to future work, again for compute reasons.

A Additional Experiments

B Implementation Details

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